

Basic Principles of Medicine 1

Module : Foundations to Medicine

Directed Learning activity

Title: Structure and function-Lipids: Part 1

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Assigned Reading

1. [Phospholipid, Glycosphingolipid, and Eicosanoid Metabolism | Lippincott® Illustrated Reviews: Biochemistry, 8e | Medical Education | Health Library \(lwwhealthlibrary.com\)](#)

Questions: 17.2 p. 218

2. Reference: Access Medicine: The Big Picture-Medical Biochemistry; Chapter 3; [LIPIDS | The Big Picture: Medical Biochemistry | AccessMedicine | McGraw Hill Medical \(oclc.org\)](#)
3. Reference: Access Medicine: Harper's Review of Biochemistry
4. Reference: <http://themedicalbiochemistrypage.org/lipids.php>

- This is a review of organic chemistry
- To enable understanding of structure and function of lipids in biological membranes
- Overview of chemical nature of medically significant lipids

Directed Learning activity

- **Two videos**
 - Fatty acids, Cholesterol, Triacylglycerol – Part 1
 - Complex lipids: Phospholipids and glycolipids – Part 2
- **Focus**
 - **Nutritional/ Medical significance**
 - **Membrane lipids**
- **The DLA also has questions and answers with brief explanation (for self-evaluation after completion of the DLA)**

Objectives: DLA: Structure and Function of Lipids

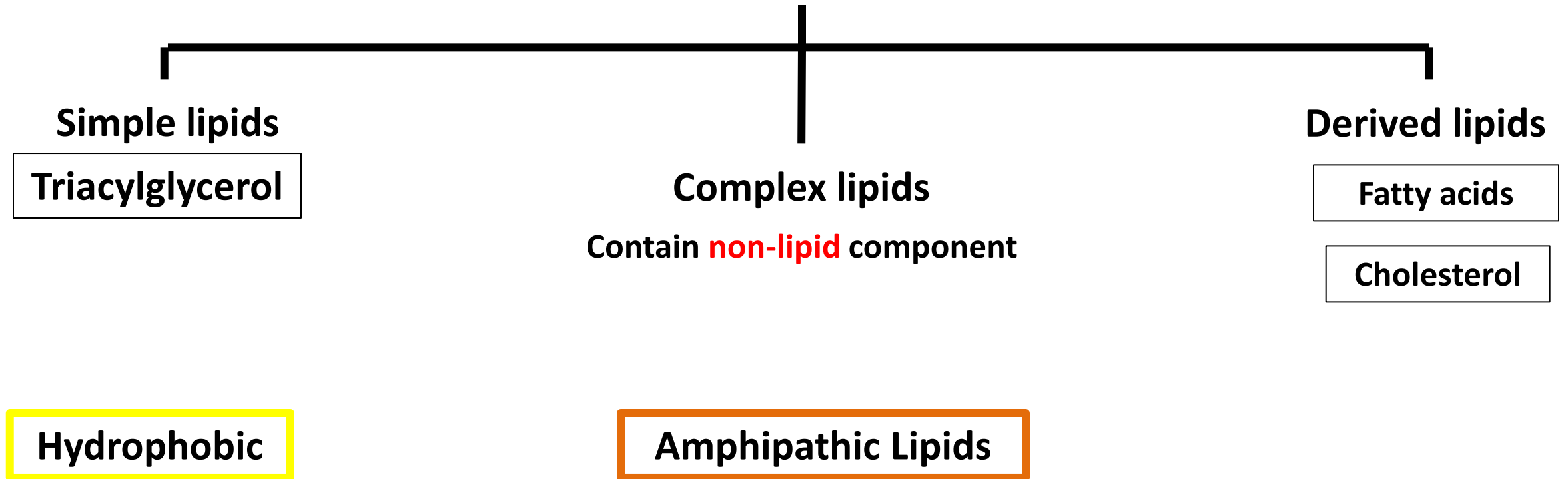
SOM.MK.I.BPM1.1.FTM.3.BCHM.0001	Review the grouping of lipids (Fatty acids, triacylglycerols, phospholipids, sphingolipids, steroids)
SOM.MK.I.BPM1.1.FTM.3.BCHM.0002	Review the significance and functions of cholesterol in humans.
SOM.MK.I.BPM1.1.FTM.3.BCHM.0003	Describe fatty acid structure. Discuss the relationship between chain length and desaturation and relate its significance to fluidity of cell membrane
SOM.MK.I.BPM1.1.FTM.3.BCHM.0004	Discuss importance of dietary essential fatty acids: linoleic acid (omega-6) and alpha-linolenic acid (omega-3)
SOM.MK.I.BPM1.1.FTM.3.BCHM.0005	Review the grouping of fatty acids into omega-6 and omega-3 families and describe the synthesis of arachidonic acid and of docosahexaenoic acid (DHA)
SOM.MK.I.BPM1.1.FTM.3.BCHM.0006	Describe structure-function relationship of triacylglycerols.
SOM.MK.I.BPM1.1.FTM.3.BCHM.0007	Distinguish between phospholipids (Sphingophospholipid and glycerophospholipids) and glycolipids with examples for each
SOM.MK.I.BPM1.1.FTM.3.BCHM.0008	Indicate composition and functions of: Glycerophospholipids (Phosphatidylcholine, phosphatidylethanolamine, phosphatidylserine, phosphatidylinositol, cardiolipin), Sphingophospholipid (Sphingomyelin), Glycolipids (Cerebrosides, globosides and gangliosides)
SOM.MK.I.BPM1.1.FTM.3.BCHM.0009	Discuss role of lung surfactant (Dipalmitoyl phosphatidylcholine/ DPPC) in respiratory distress syndrome

Fatty acids, Cholesterol and Triacylglycerol

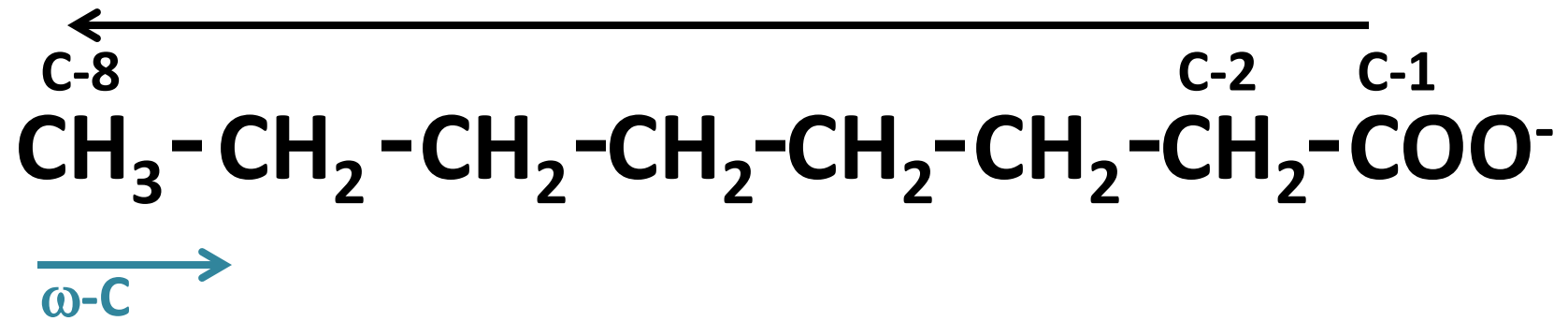
Objectives - Part 1

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Classification of lipids



Fatty acids



Classification of fatty acids

(Based on presence of **double bonds**)

Saturated fatty acids

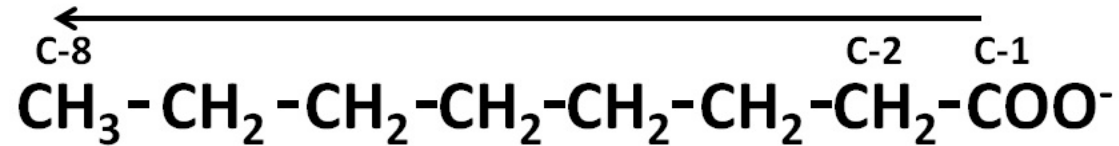
Long chain fatty acids
(12-20 C atoms)
Eg: Palmitic acid, stearic acid

Unsaturated fatty acids

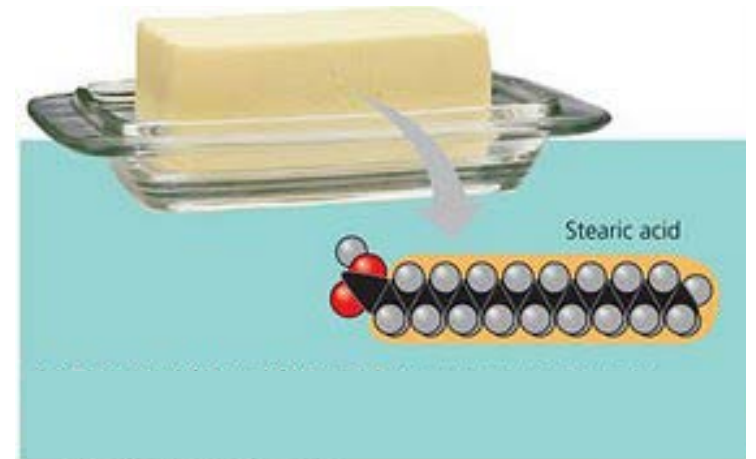
Monounsaturated
(MUFA)

Polyunsaturated fatty
acids (PUFA)

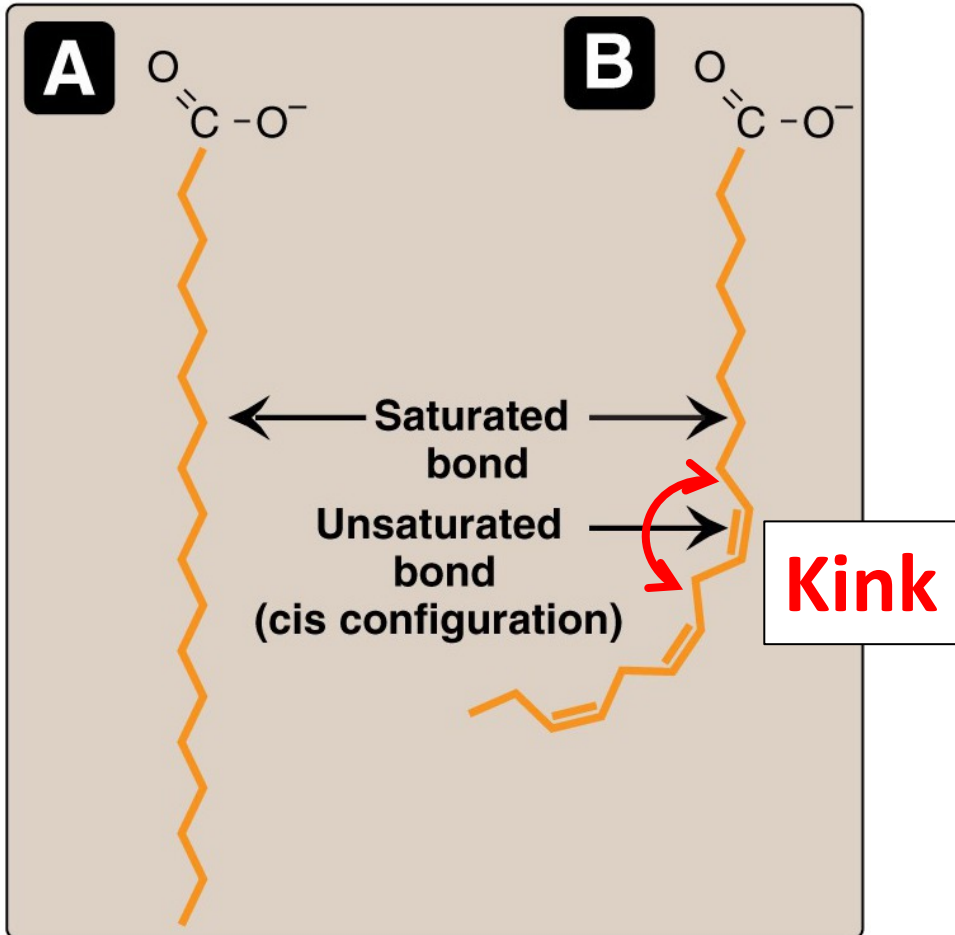
Saturated fatty acids



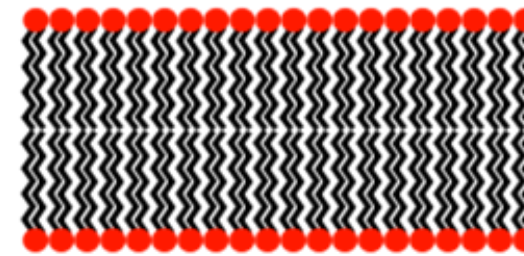
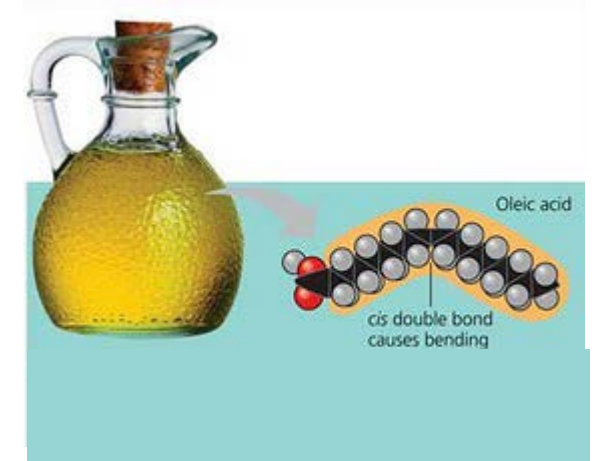
- Longer chain length, close packing and lower mobility in membranes
- Lipids with saturated fatty acids are solid
- **Palmitic acid (16-C)** and stearic acid (18 C)
- Longer chain length rich in myelin
- **Nutrition Pearl:** Limit dietary saturated fatty acid content for cardiovascular health



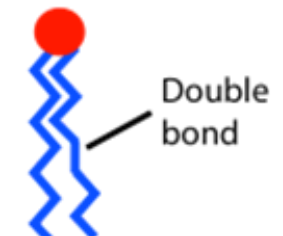
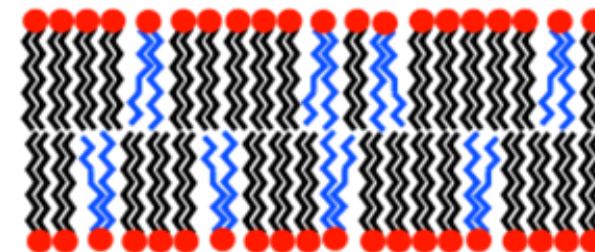
Unsaturated fatty acids



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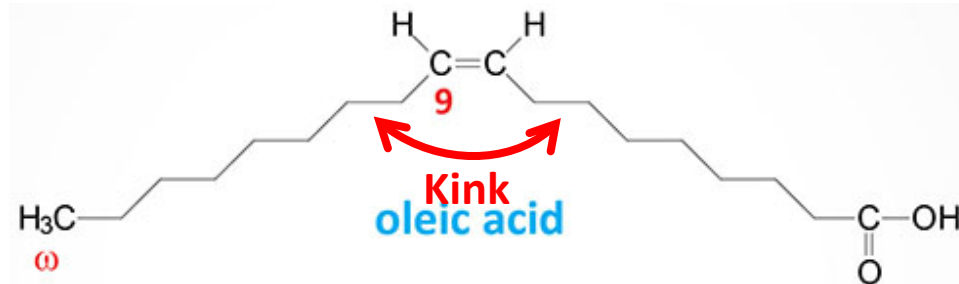


Saturated lipids only



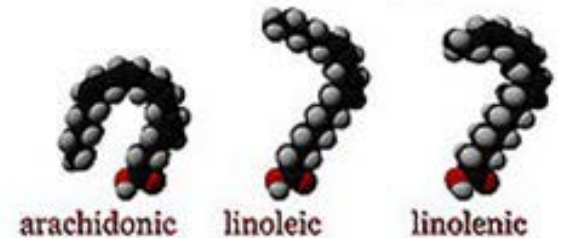
Monounsaturated fatty acid

- Oleic acid (18:1)
- Olive oil rich in oleic acid (ω -9)
- 'Mediterranean' diet



Polyunsaturated fatty acids (PUFA)

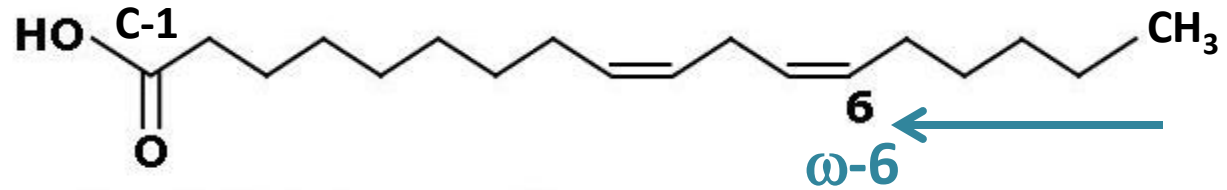
- More number of double bonds
 - More **fluidity** of membranes
 - Multiple kinks prevent close packing
- Two families:
 - ω -3: **α -Linolenic acid**; Eicosapentaenoic acid and DHA
 - ω -6: **Linoleic acid** and arachidonic acid
- Nutrition Pearl:
 - **More ω -3** and less ω -6 fatty acids;
 - Lower dietary ω -6: ω -3 fatty acid ratio



PUFA: ω -6 fatty acids



Essential fatty acid

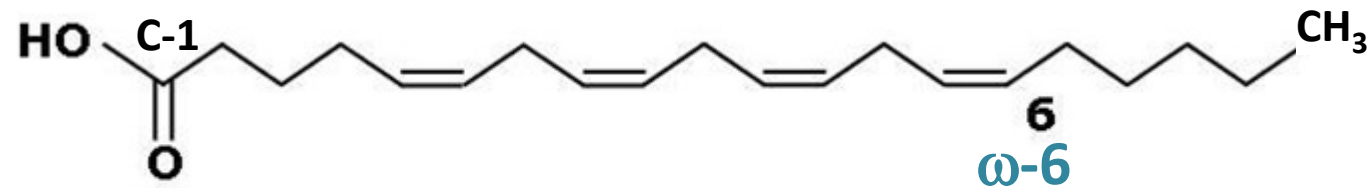
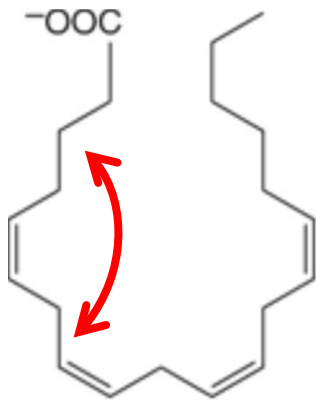


Linoleic acid (LA, C18:2, omega-6)

Desaturation (Addition of a double bonds)

Elongation (Addition of 2 C-atoms)

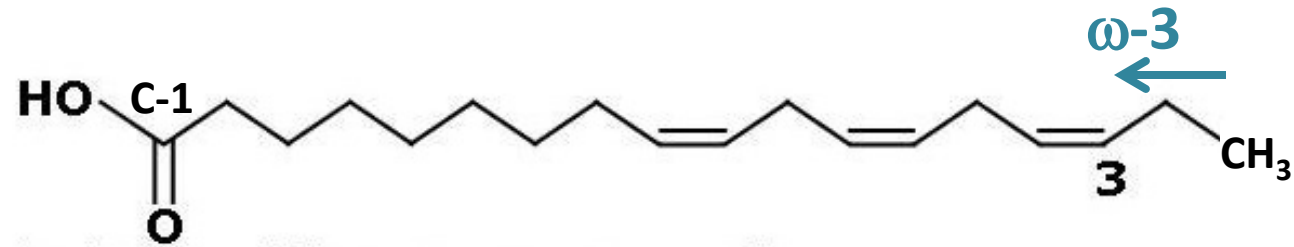
Kink



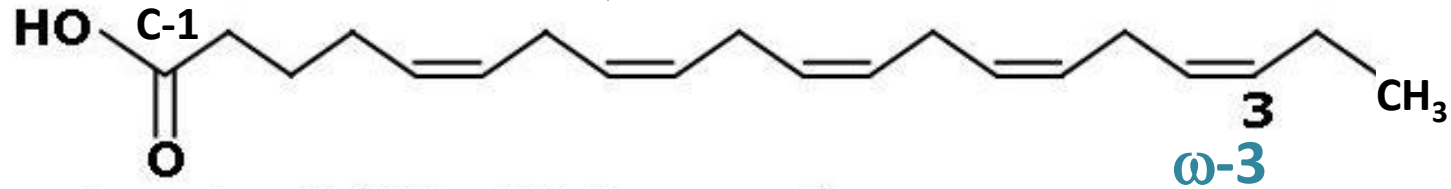
Arachidonic acid (AA, C20:4, omega-6)

PUFA: ω -3 fatty acids

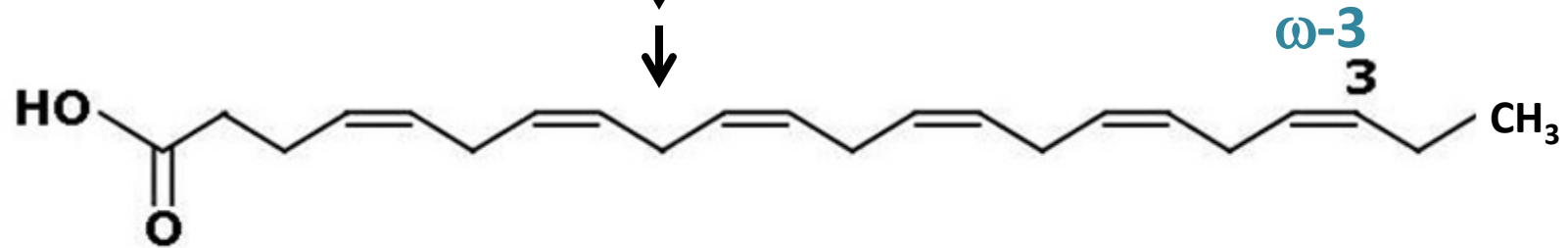
Essential fatty acid



Alpha-linolenic acid (ALA, C18:3, omega-3)



Eicosapentaenoic acid (EPA, C20:5, omega-3)



Docosahexaenoic acid (DHA, C22:6, omega-3)

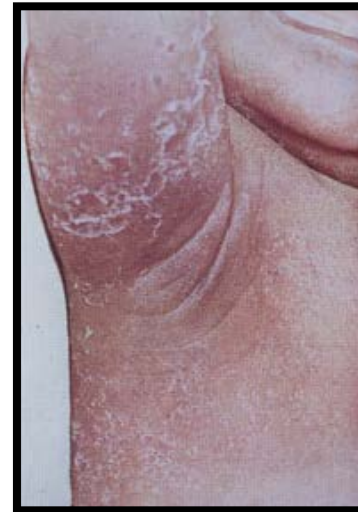
ω -3 fatty acids: α -Linolenic acid, EPA, docosahexaenoic acid (DHA)



- Eicosanoids from EPA are cardioprotective
- DHA important for neural/ brain development and for vision
- **Presence of which fatty acid makes membrane most fluid???**
- **Name the omega-3 and omega-6 fatty acids**

Essential fatty acids

- Humans **cannot add double bond after C-9** (starting from carboxyl end)
- **Linoleic acid** (ω -6) and **α -linolenic acid** (ω -3) dietary **essential fatty acids**
 - Cannot be synthesized
 - Must be provided in diet
- Nutritional deficiency: Scaly dermatitis, rough and dry skin, loss of hair



Fatty acids with chain lengths of 4 to 10 carbons are found in significant quantities in milk.

Structural lipids and triacylglycerols contain primarily fatty acids of at least 16 carbons.

COMMON NAME	STRUCTURE
Formic acid	1
Acetic acid	2:0
Propionic acid	3:0
Butyric acid	4:0
Capric acid	10:0
Palmitic acid	16:0
Palmitoleic acid	16:1(9)
Stearic acid	18:0
Oleic acid	18:1(9)
Linoleic acid	18:2(9,12)
α -Linolenic acid	18:3(9,12,15)
Arachidonic acid	20:4(5, 8,11,14)
Lignoceric acid	24:0
Nervonic acid	24:1(15)

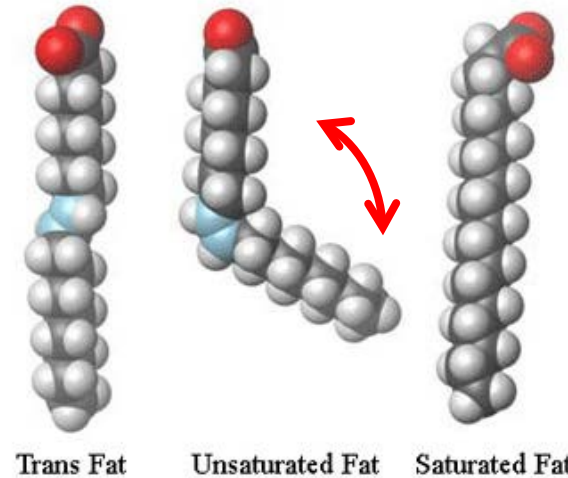
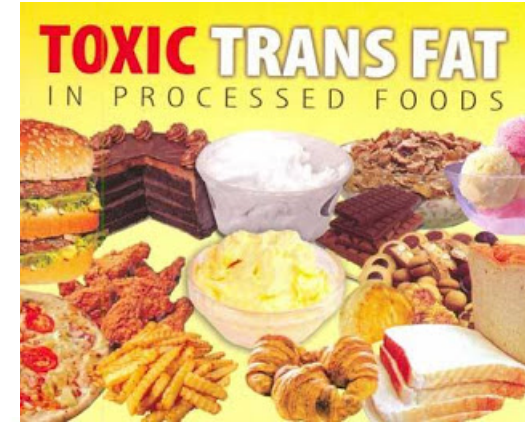
Precursor of prostaglandins

Essential fatty acids

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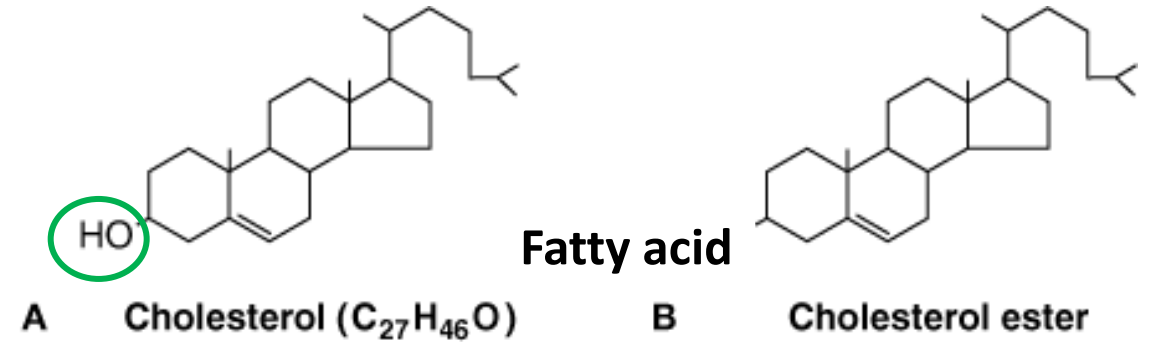
Trans fatty acids

- ‘**trans**’ conformation; **NO** kink in structure
- Limited mobility (Act like saturated fatty acids)
(compare unsaturated fatty acids with ‘cis’ bonds)
- **Nutrition Pearl:** NO ‘trans’ fatty acid
 - Increase LDL cholesterol and **risk of heart disease**
- American Heart Association (AHA): Trans fats
[http://www.heart.org/HEARTORG/HealthyLiving/HealthyEating/Nutrition/Trans-Fats UCM 301120 Article.jsp#.V2gNApevzd4](http://www.heart.org/HEARTORG/HealthyLiving/HealthyEating/Nutrition/Trans-Fats_UCM_301120_Article.jsp#.V2gNApevzd4)
- Partially (artificially) hydrogenated lipids like **margarine**, fried foods, baked goods...

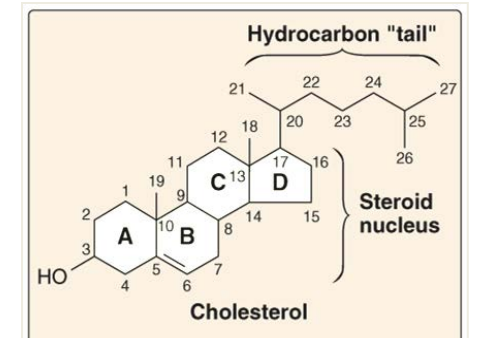


Nutrition Facts		Amount/Serving	%Daily Value*
Total Fat 14g			22%
Saturated Fat 4g			20%
Trans Fat 3.5g			
Cholesterol 60mg			20%
Sodium 200mg			8%
Vitamin A 0%		Vitamin C 0%	

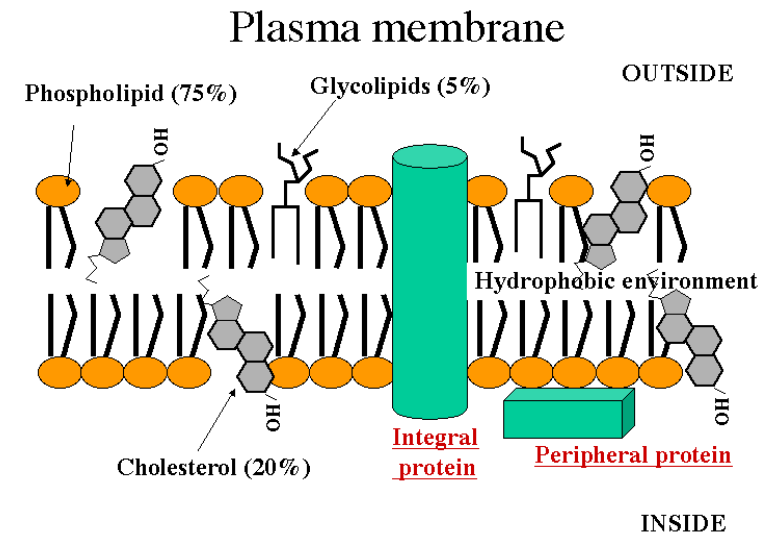
Cholesterol



- 27-C atom
- Cyclopentanoperhydrophenanthrene (**steroid**) ring
- **Amphipathic lipid**
 - Hydrophilic **hydroxyl (OH) group**
 - Hydrophobic steroid ring
- Regulation of membrane fluidity



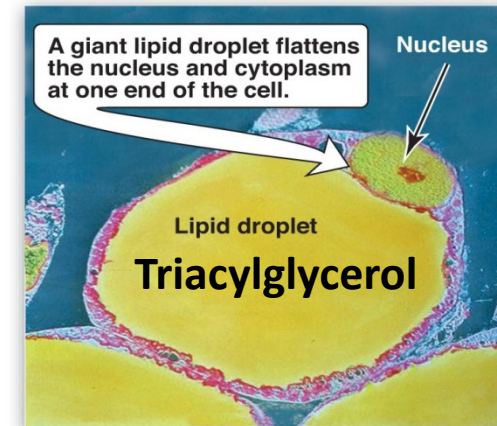
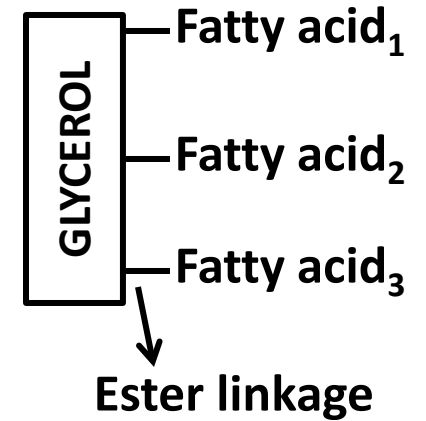
Functions of cholesterol



- **Cell/ plasma membrane (both leaflets)**
- Precursor of **bile acid/ bile salt synthesis** in liver
 - Efficient intestinal digestion and absorption of dietary lipids
- Precursor of **steroid hormones**:
 - Cortisol and aldosterone in adrenal cortex
 - Sex steroids: Estrogen and progesterone in ovaries; Testosterone in testes
- Precursor for synthesis of **vitamin D** in skin

Simple lipids: Triacylglycerol (Triglyceride)

- About 90-95% of dietary lipids are TAGs
 - Fats (butter...)
 - Oils
- Storage form of energy in adipose tissue
 - **Hydrophobic**
 - 70-kg adult stores 15 kg as adipose TAG (fat)
 - Sources of energy during fasting
- **NOT components of membrane**



Adipocyte with stored TAG

Derived lipids: Examples and functions

Lipid	Examples	Function/ Significance
Saturated fatty acids	Palmitic acid (C16), Stearic acid (C18)	Components of TAG, phospholipids and glycolipids (Decrease membrane fluidity)
Monounsaturated fatty acids	Oleic acid (C18)	Components of TAG, phospholipids and glycolipids
Polyunsaturated fatty acids	Omega-3 and omega-6 fatty acids	Components of TAG, phospholipids and glycolipids (Increase membrane fluidity)
Omega-3 fatty acids	Alpha-linolenic acid , Eicosapentaenoic acid, Docosahexaenoic acid	Components of TAG, phospholipids and glycolipids (Increase membrane fluidity)
Omega-6 fatty acids	Linoleic acid , Arachidonic acid	Components of TAG, phospholipids and glycolipids (Increase membrane fluidity)
Essential fatty acids	Alpha-linolenic acid, Linoleic acid	Components of TAG, phospholipids and glycolipids (Increase membrane fluidity)
Cholesterol	--	Cell membrane (both leaflets); Precursor of steroid hormones, vit D and bile acids

Take home points

- List the following:
 - Saturated, monounsaturated and polyunsaturated fatty acids
 - Omega-6 and omega-3 fatty acids
 - Essential fatty acids
- Relate fatty acid to membrane fluidity:
 - Chain length
 - Double bonds
 - cis or trans double bonds
- Please email shupadhya@sgu.edu for questions



Thank You!!